

When Does Temporal Validity Help Agent Memory? A Practical Evaluation Across Memory Regimes

Adarsh Agrawal
Senior Member, IEEE

Abstract

Long-horizon agents need to update their beliefs when new evidence makes old memory stale. We test a training-free layer that revises stored beliefs, ranks evidence with temporal decay, and can decline an answer when support is weak. We compare up to twelve policies across LongMemEval drift QA, LME-drift, an 812-item changed-fact derivative, and 451 LongMemEval-V2 questions under a text-only, oracle-trajectory protocol (V2-perQ). We report answer quality, coverage, and the conditional error rate among committed answers, denoted SF as a stale-fact shorthand on these drift-focused populations. On held-out questions, the decay-based hybrid lowers SF by 0.017, 0.052, and 0.081 across the three regimes. Utility is unresolved on the small split but improves by 0.034 and 0.047 on the larger tests; V2-perQ answer quality also improves by 0.032. Held-out V2-perQ ablations identify temporal decay as the productive component, while hard gating loses coverage. The cross-benchmark pattern is consistent with greater benefit under heavier stale-evidence burden, but it is not a causal estimate of memory horizon.

Introduction

Long-term memory is becoming a standard part of language-agent systems. Recent work organizes experience through evolving notes, graphs, selective extraction, and retrieval tuned for continual learning [1–4]. The hard part is not only remembering more. An agent also has to recognize when an old statement has been replaced. Recent benchmarks show that updating, forgetting, and multi-session consistency remain difficult even when ordinary recall looks strong [5–8]. Related 2026 benchmarks extend this view to interdependent agent tasks and explicit forgetting [12, 13]. Newer systems address these failures with explicit memory operations, revocation, decay, and temporal reasoning [14–18].

We study a training-free temporal-validity layer that revises stored beliefs, tracks where they came from, ranks surviving evidence with temporal decay, and can abstain when support is weak. The main question is diagnostic: when the system improves, is the gain coming from better retrieval, hard deletion, or refusing more questions? We compare all policies with the same answer-quality, stale-fact, coverage, confabulation, and utility measures. A small factorial and focused ablations separate the retrieval and gating effects.

Contributions. We make three contributions. First, we evaluate temporal-memory policies on held-out questions using one set of metrics and a two-part requirement: answer quality

should not meaningfully fall while stale answers decrease. Second, we separate decay-based retrieval from lifecycle gating and test both on held-out V2-perQ questions. Third, we show a benchmark-dependent pattern across LongMemEval, LME-drift, and V2-perQ [5, 23]. The pattern and two supporting within-regime analyses are consistent with a stale-evidence explanation, while raw-context replay remains a strong comparison.

Approach

Temporal-validity layer. Each BeliefRecord stores a value, source, reliability estimate, valid-time interval, transaction-time interval, and lifecycle status. REVISE closes the old version before opening its replacement, while REVOKE closes a record and removes it from retrieval. Surviving records are ranked using source reliability and temporal decay. A final status check prevents most revoked facts from returning through semantic similarity. We call the variants hybrid (decay plus selective gating), decay-only, gate-only, and full layer (all lifecycle operations).

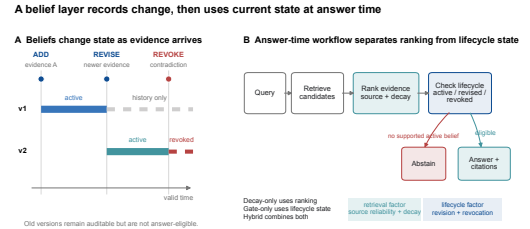


Figure 1: Belief updates and answer-time control. Ranking selects evidence; lifecycle state controls answer eligibility.

Metrics and decision rule. For question i , model m , and policy s , let a_{smi} indicate abstention and c_{smi} indicate a correct committed answer. We aggregate models as repeated measures on the question and define answer quality $AQ = \mathbb{E}[c]$, coverage $CV = \mathbb{E}[1 - a]$, and SF = $\Pr(c=0 \mid a=0)$, yielding $AQ = CV(1 - SF)$. SF is exactly the conditional error rate among committed answers. We retain “stale-fact rate” as a functional shorthand because stale evidence is the dominant audited error type on these drift-focused populations; Appendix reports the breakdown. Confabulation is the answer rate on unanswerable traps.

For each question, define the raw deployment score

$$R_i = \frac{1}{3} \sum_m [c_{im} - 5w_{im} - 0.1a_{im}] .$$

Here $w_{im} = (1 - a_{im})(1 - c_{im})$. We report normalized utility $U = \mathbb{E}[(R_i + 5)/6] \in [0, 1]$. This affine transformation preserves every paired difference, $\Delta U = \Delta R/6$. Appendix Table 6 gives the outcome decomposition.

Comparisons. Thresholds and decay rates are chosen on a tuning split, leaving the test questions untouched. We treat answer quality as maintained when the hybrid is within $\delta=0.05$ of RAG, and call the two-part result successful only when stale-fact rate also falls. Utility is tested directly at the cost point above. The margin was fixed before held-out evaluation and bounds normalized utility loss between 0.009 and 0.050 per item at the target costs. Within each benchmark, fixed-sequence gatekeeping tests AQ non-inferiority (one-sided $\alpha=0.025$), then SF superiority, then utility superiority (both two-sided $\alpha=0.05$); a later test is interpreted only if every earlier test passes. This controls the within-benchmark familywise error rate. The three benchmarks address separate populations and are treated as separate families. Confidence intervals resample questions while keeping the three model responses together.

Benchmarks. LongMemEval contributes all 199 answerable drift questions and 30 unanswerable traps from its s split; the answerable items split 101/98 and the traps split 15/15. LME-drift derives 812 changed-fact questions from 500 source sessions and splits by source session (406/406); it is distinct from official LongMemEval-M. V2-perQ evaluates all 451 LongMemEval-V2 questions using only the question-generating trajectory (mean 47K, maximum 215K tokens), rather than the official multi-trajectory haystack, and splits questions by trajectory (226/225). This oracle reduction holds trajectory selection fixed, isolating temporal ranking and lifecycle control from cross-trajectory retrieval. Screenshots are excluded and only trajectory text is processed. Appendix Table 1 gives full construction details. Every policy uses three frozen models at temperature zero. RAG retrieves the top ten passages by cosine similarity from a Titan Text Embeddings V2 index with a 4096-token evidence budget; Appendix gives the complete specification. Raw context replays the per-question history.

Results

Held-out effects across memory regimes. Figure 2 reports paired held-out contrasts. On LongMemEval s, hybrid and RAG have equal answer quality (0.000, CI $[-0.048, 0.048]$); the one-sided lower bound exceeds $-\delta$ ($p_{NI}=0.020$). The hybrid lowers stale-fact rate (-0.017 , CI $[-0.033, -0.001]$, $p=0.013$), so it meets the two-part answer-quality and stale-fact criterion. Its utility difference is positive but unresolved ($+0.018$, CI $[-0.015, 0.051]$, superiority $p=0.14$).

On LME-drift, the hybrid remains non-inferior in answer quality ($+0.008$, CI $[-0.014, 0.030]$, $p_{NI}<0.001$), lowers stale-fact rate (-0.052 , CI $[-0.072, -0.032]$, $p<0.001$), and improves utility ($+0.034$, CI $[0.009, 0.059]$, $p=0.004$). On V2-perQ it is superior to RAG in answer quality ($+0.032$, CI $[0.005, 0.059]$, $p=0.024$), stale-fact rate (-0.081 , CI $[-0.112, -0.050]$, $p<0.001$), and utility ($+0.047$, CI $[0.012, 0.082]$, $p=0.008$).

On the 15 held-out LongMemEval traps, hybrid commits an answer on 0.40 versus RAG’s 0.67; the wide intervals make this directional evidence (Appendix Table 8).

Coverage and answered accuracy across policies. On V2-perQ, decay-only has the highest answer-quality point estimate (0.28), followed by hybrid (0.27), raw context (0.26), and RAG (0.24). On the held-out split, decay-only improves over RAG in AQ ($+0.038$, CI $[0.008, 0.068]$) and SF (-0.059 , CI $[-0.091, -0.027]$). Gate-only lowers SF further but reduces AQ (-0.058 , CI $[-0.089, -0.027]$). Hybrid and decay-only are unresolved in AQ (-0.006 , CI $[-0.031, 0.019]$), while hybrid exceeds gate-only by 0.090. Thus the V2-perQ held-out ablation identifies decay as the productive component; the smaller LongMemEval factorial remains exploratory.

Raw context, stability, and long context. Raw-context replay remains a strong comparator. Hybrid-minus-raw AQ is -0.040 on LongMemEval (CI $[-0.081, 0.001]$, $p=0.056$), -0.002 on LME-drift (CI $[-0.024, 0.020]$, $p=0.43$), and $+0.010$ on V2-perQ (CI $[-0.019, 0.039]$, $p=0.25$); none is statistically resolved. The claim is therefore a win over fixed-memory RAG in longer regimes, not universal dominance over complete-history replay.

Panel B of Figure 2 shows that the small split is unstable: raw context is the utility winner in 81% of 200 question-clustered resplits and hybrid in 15%. On LME-drift, this reverses to 31% raw and 62% hybrid; on V2-perQ, 18% raw and 78% hybrid. These resplits are sensitivity evidence rather than confirmatory tests. On the 47 V2-perQ histories exceeding 150K context tokens, hybrid reaches 0.23 answer quality versus RAG’s 0.18 and a 0.43 SF versus 0.58. In an exploratory median split by stale-evidence density, hybrid-minus-RAG AQ is $+0.011$ (CI $[-0.024, 0.046]$) in the low-density half and $+0.053$ (CI $[0.018, 0.088]$) in the high-density half. Because benchmark and density groups differ on other dimensions, these patterns are consistent with, but do not establish, a horizon effect.

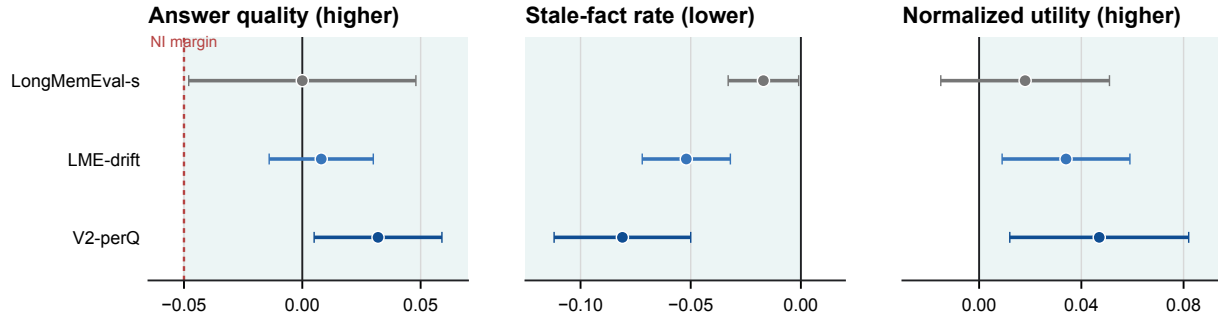
Cost and failure cases. Relative to RAG, decay-only adds 11ms and 82 prompt tokens per query, hybrid adds 23ms and 142 tokens, and the full layer adds 47ms and 218 tokens. Raw-context inputs average 12K, 34K, and 47K tokens across the three regimes (maximum 215K), with no truncation or context failures, but cost 3–8 $\times$ more than RAG. The store reaches 8ms p95 lookup at 50K beliefs. Two annotators label 78% of 247 hybrid citations as direct support versus 71% of 231 RAG citations ($\kappa=0.74$). Appendix reports failure rates and denominators.

What this means in practice. The scope is factual drift in text memory. Raw context remains attractive while replay is affordable; decay becomes useful as stale evidence accumulates, and selective gating is justified when wrong answers cost more than abstentions. Appendix states the remaining scope boundaries.

Held-out effects and split sensitivity across three memory regimes

Top: point estimates with 95% CIs. Bottom: sensitivity to split composition.

A Held-out effect: hybrid - RAG



B Winner share across 200 resplits

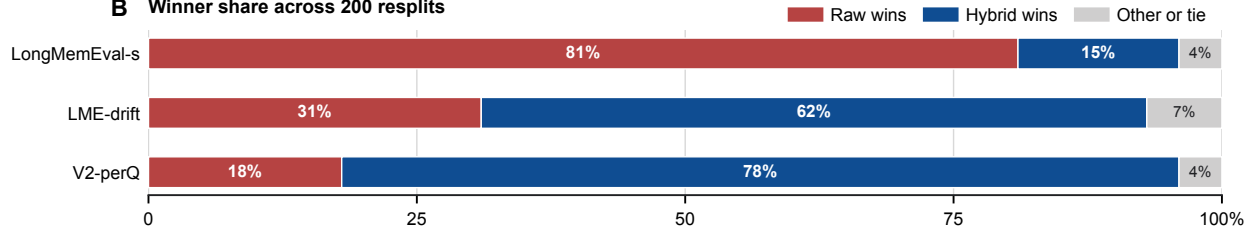


Figure 2: Held-out effects and split sensitivity across memory regimes. Panel A shows paired differences with 95% CIs and the AQ non-inferiority margin. Panel B shows utility winners across 200 retuned sensitivity resplits.

Coverage and answered accuracy across policies

V2-perQ full pool; dotted contours satisfy $AQ = CV(1 - SF)$

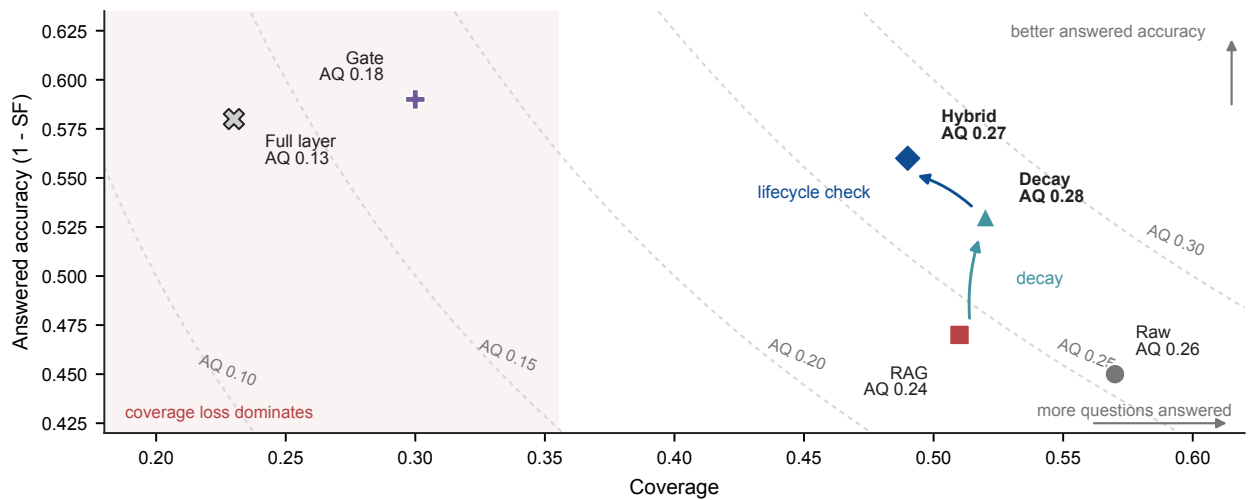


Figure 3: Coverage and answered accuracy on V2-perQ. Full-pool descriptive values; dotted curves satisfy $AQ = CV(1 - SF)$. Decay-based policies have higher answered accuracy with modest coverage reduction, while gating policies cover fewer questions.

Conclusion

Temporal validity is useful but not a universal hard gate. Across three benchmarks, hybrid lowers committed-answer error relative to RAG, and the larger tests support normalized-utility gains. Held-out V2-perQ ablations identify decay-based ordering as the productive component; gate-only policies trade coverage for fewer errors. Horizon-related patterns are consistent but not causal. Deployment should choose among raw context, decay-only retrieval, and selective gating based on replay and error costs.

References

- [1] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-MEM: Agentic memory for LLM agents. *Advances in Neural Information Processing Systems*, 2025.
- [2] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: A temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956*, 2025.
- [3] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.
- [4] Bernal Jiménez Gutiérrez, Yiheng Shu, Weijian Qi, Sizhe Zhou, and Yu Su. From RAG to memory: Non-parametric continual learning for large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 21497–21515, 2025.
- [5] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations*, 2025.
- [6] Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating memory in LLM agents via incremental multi-turn interactions. *arXiv preprint arXiv:2507.05257*, 2025.
- [7] Ding Chen, Simin Niu, Kehang Li, Peng Liu, Xiangping Zheng, Bo Tang, Xinchu Li, Feiyu Xiong, and Zhiyu Li. HaluMem: Evaluating hallucinations in memory systems of agents. *arXiv preprint arXiv:2511.03506*, 2025.
- [8] Haoran Tan, Zeyu Zhang, Chen Ma, Xu Chen, Quanyu Dai, and Zhenhua Dong. MemBench: Towards more comprehensive evaluation on the memory of LLM-based agents. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 19336–19352, 2025.
- [9] Shashank Indukuri and Adarsh Agrawal. Grounded optimization: A layered engineering framework for reducing LLM hallucination in automated personal document rewriting. *arXiv preprint arXiv:2607.01457*, 2026.
- [10] Amazon AGI Intelligence. Amazon nova premier: Technical report and model card. Technical report, Amazon, 2025.
- [11] Bertie Vidgen, Adarsh Agrawal, Ahmed M. Ahmed, Victor Akinwande, Noura Al-Nuaimi, Nouf Alfaraj, et al. Introducing v0.5 of the AI safety benchmark from ML-Commons. *arXiv preprint arXiv:2404.12241*, 2024.
- [12] Zexue He, Yu Wang, Churan Zhi, Yuanzhe Hu, Tzu-Ping Chen, Lang Yin, Ze Chen, Tong Arthur Wu, Siru Ouyang, Zihan Wang, Jiaxin Pei, Julian McAuley, Yejin Choi, and Alex Pentland. MemoryArena: Benchmarking agent memory in interdependent multi-session agentic tasks. *arXiv preprint arXiv:2602.16313*, 2026.
- [13] Md Nayem Uddin, Kumar Shubham, Eduardo Blanco, Chitta Baral, and Gengyu Wang. From recall to forgetting: Benchmarking long-term memory for personalized agents. *arXiv preprint arXiv:2604.20006*, 2026.
- [14] Hanxiang Chao, Yihan Bai, Rui Sheng, Tianle Li, and Yushi Sun. STALE: Can LLM agents know when their memories are no longer valid? *arXiv preprint arXiv:2605.06527*, 2026.
- [15] Xixuan Hao, Zeyu Zhang, Zehao Lin, Yihang Sun, Ziliang Guo, Xichong Zhang, Yuxuan Liang, Feiyu Xiong, and Zhiyu Li. MemOps: Benchmarking lifecycle memory operations in long-horizon conversations. *arXiv preprint arXiv:2607.12893*, 2026.
- [16] Yan Zhou, Yue Ouyang, Kaiyang Zheng, and Suncheng Xiang. TEPA: Revoking stale memories for conflict-robust language agents. *arXiv preprint arXiv:2608.07429*, 2026.
- [17] Xiaoyang Li, Yiqi Wang, Haohui Lu, Zhi Chen, Mo Li, Pingan Song, Mingkai Zheng, and Taotao Cai. MemTX: Transactional belief commit for stateful agent memory. *arXiv preprint arXiv:2607.23929*, 2026.
- [18] Muntaser Syed, Marius Silaghi, Sheikh Abujar, and Sharun Akter. Chronofy: A temporal-logical decay architecture for information validity in time-aware retrieval-augmented generation. *arXiv preprint arXiv:2607.20560*, 2026.
- [19] Adarsh Agrawal and Rahul Suresh Babu. Incident memory: Training-free operational memory through sequential pattern mining and velocity-stratified retrieval. *arXiv preprint arXiv:2609.01616*, 2026.
- [20] Amazon AGI Intelligence. Amazon nova sonic: Technical report and model card. Technical report, Amazon, 2025.
- [21] Adarsh Agrawal and J. Li. Mitigating bias in AI using Debias-GAN. NVIDIA GTC Conference, 2022.
- [22] Adarsh V. Agrawal and Nikhil Garg. Time series forecasting of producer price index, using ARIMA. *International Journal of Science and Research*, 7(7):1019–1024, 2018.

- [23] Di Wu, Zixiang Ji, Asmi Kawatkar, Bryan Kwan, Jia-Chen Gu, Nanyun Peng, and Kai-Wei Chang. LongMemEval-V2: Evaluating long-term agent memory toward experienced colleagues. *arXiv preprint arXiv:2605.12493*, 2026.
- [24] Adarsh Agrawal and Shashank Indukuri. Schema-first retrieval: Embedding catalogs for natural language analytics. *arXiv preprint arXiv:2606.28387*, 2026.
- [25] Amazon AGI Intelligence. Amazon nova multimodal embeddings: Technical report and model card. Technical report, Amazon, 2025.
- [26] Bertie Vidgen, Adarsh Agrawal, Ahmed M. Ahmed, Victor Akinwande, Noura Al-Nuaimi, Nouf Alfaraj, et al. Introducing v0.5 of the AI safety benchmark from MLCommons, 2024. <https://arxiv.org/abs/2404.12241>, 2024.
- [27] Wei Feng, Adarsh Agrawal, Haifeng Ling, Erik Blasch, Eric Adiles-Cruz, Paul T. Schrader, and Jun Wei. DDDAS probability learning for natural disaster change detection. In *International Conference on Dynamic Data Driven Applications Systems*, pages 90–99, 2024.
- [28] Rahul Suresh Babu and Adarsh Agrawal. Self-healing agentic orchestrators for reliable tool-augmented large language model systems. *arXiv preprint arXiv:2606.01416*, 2026.
- [29] Amazon AGI Intelligence. Amazon nova 2: Multimodal reasoning and generation models. Technical report, Amazon, 2025.
- [30] Amazon AGI Intelligence. The amazon nova family of models: Technical report and model card. Technical report, Amazon, 2024.
- [31] Bertie Vidgen, Adarsh Agrawal, Ahmed M. Ahmed, Victor Akinwande, Noura Al-Nuaimi, Nouf Alfaraj, et al. Others. 2024. introducing v0.5 of the AI safety benchmark from MLCommons. *arXiv preprint arXiv:2404.12241*, 2024.
- [32] Bertie Vidgen, Adarsh Agrawal, Ahmed M. Ahmed, Victor Akinwande, Noura Al-Nuaimi, Nouf Alfaraj, et al. How to construct AI safety benchmarks, n.d. Technical article.
- [33] Caili Yu, Yiqi Wang, Jiaqi Zhang, Yiqun Duan, Mingkai Zheng, Zhangkai Wu, Kaize Shi, and Taotao Cai. From faulty memories to corrected actions: Dependency-guided rollback repair for memory-augmented agents. *arXiv preprint arXiv:2608.10502*, 2026.

Benchmark provenance and split accounting

Table 1: Benchmark provenance. Trap items are evaluated separately from answerable questions.

Benchmark	Protocol	Content source	Answerable	Traps	Time / test	Split unit
LongMemEval s	Standard	Full s-split histories	199	30	101+15 / 98+15	question-ID hash
LME-drift	Standard	Full source-session histories	812	0	406 / 406	source session
V2-perQ	Oracle trajectory	Question-generating trajectory only	451	0	226 / 225	trajectory-ID hash

The 30 LongMemEval traps split 15/15 and measure confabulation separately. LME-drift is constructed from 500 LongMemEval source sessions. We identify entity-attribute claims that are stated and later updated or contradicted, generate one current-value question per changed fact, and retain one to three questions per contributing session. The result has 812 questions (mean 1.62 per session); all questions from a source session stay in one fold. This derivative is distinct from official LongMemEval-M.

LME-drift extraction. We use three stages. A model first lists entity-attribute pairs whose values change within each session, then generates a current-value question for each candidate. Rule checks reject unchanged values, a second pass flags ambiguity, and a manual spot-check rejects remaining errors. Of 923 candidates, 812 remain: 67 false changes, 29 ambiguous questions, and 15 manual rejects are removed. Splitting occurs only after validation and keeps every source session in one fold.

V2-perQ protocol. Official LongMemEval-V2 evaluates retrieval over haystacks containing many web-agent trajectories. V2-perQ deliberately uses oracle trajectory selection: every policy receives only the trajectory that generated the question. This is a strictly easier task than the official full-haystack protocol. It controls trajectory retrieval so policy differences reflect temporal ranking and lifecycle gating within the same evidence. The 47K mean and 215K maximum therefore describe single question-trajectory pairs, not the aggregate V2 haystack. Questions are grouped by `trajectory_id`; the split audit verifies zero ID overlap.

V2 also contains multimodal trajectories. Our projection retains textual actions, page text, and URL paths, but excludes screenshots and other visual elements. We retain all 451 questions and apply this projection identically to every policy. This preserves a common input condition, but questions that require visual evidence can depress absolute accuracy and may add noise to policy contrasts. We do not claim performance on V2’s full-haystack or multimodal task.

Complete policy tables

Table 2: LongMemEval s full-pool metrics. Confidence intervals are question-clustered 95% bootstrap intervals.

Policy	AQ	AQ CI	SF	SF CI
raw context	0.31	[0.27, 0.35]	0.49	[0.45, 0.53]
vanilla RAG	0.28	[0.24, 0.32]	0.49	[0.45, 0.53]
hybrid	0.27	[0.23, 0.31]	0.48	[0.44, 0.52]
full layer	0.16	[0.13, 0.19]	0.50	[0.46, 0.54]
Chronofy [†]	0.30	[0.26, 0.34]	0.48	[0.44, 0.52]
TEPA [†]	0.15	[0.12, 0.18]	0.48	[0.44, 0.52]
STALE [†]	0.06	[0.04, 0.08]	0.29	[0.23, 0.35]
last-write-wins	0.23	[0.19, 0.27]	0.51	[0.47, 0.55]
recency-only	0.25	[0.21, 0.29]	0.50	[0.46, 0.54]
decay-only	0.28	[0.24, 0.32]	0.48	[0.44, 0.52]
gate-only	0.18	[0.15, 0.21]	0.47	[0.43, 0.51]
oracle	0.42	[0.38, 0.46]	0.32	[0.28, 0.36]

[†]Paper-inspired training-free reduction, not official code. These comparisons measure our implementations and do not establish intrinsic limits of the published methods.

Table 3: Complete full-pool policy results for the larger benchmarks.

Policy	LME-drift			V2-perQ		
	AQ	SF	CV	AQ	SF	CV
raw context	0.29	0.52	0.60	0.26	0.55	0.57
vanilla RAG	0.26	0.51	0.54	0.24	0.53	0.51
hybrid	0.29	0.44	0.51	0.27	0.44	0.49
full layer	0.16	0.45	0.29	0.13	0.42	0.23
Chronofy [†]	0.27	0.50	0.53	0.24	0.52	0.50
TEPA [†]	0.14	0.45	0.25	0.12	0.44	0.21
STALE [†]	–	–	–	0.05	0.26	0.07
decay-only	–	–	–	0.28	0.47	0.52
gate-only	–	–	–	0.18	0.41	0.30
last-write-wins	–	–	–	0.21	0.53	0.44
oracle	–	–	–	0.42	0.28	0.59

Held-out comparisons and sensitivity

Table 4: Hybrid minus RAG on held-out questions. AQ non-inferiority uses $\delta=0.05$; utility p -values are two-sided superiority tests.

Benchmark	AQ difference [95% CI]	SF difference [95% CI]	Utility difference [95% CI]	Utility p
LongMemEval s	0.000 [−0.048, 0.048]	−0.017 [−0.033, −0.001]	0.018 [−0.015, 0.051]	0.14
LME-drift	0.008 [−0.014, 0.030]	−0.052 [−0.072, −0.032]	0.034 [0.009, 0.059]	0.004
V2-perQ	0.032 [0.005, 0.059]	−0.081 [−0.112, −0.050]	0.047 [0.012, 0.082]	0.008

Table 4 contains the primary held-out comparisons. The extra policy comparisons, alternative margins, factorial effects, and temporal split are exploratory. Panel B of Figure 2 shows how often policy selection changes across 200 question-clustered resplits.

Table 5: Sensitivity to the AQ non-inferiority margin.

Benchmark	$\delta=0.05$	$\delta=0.03$	$\delta=0.02$
LongMemEval s	NI ($p=0.020$)	not NI ($p=0.14$)	not NI ($p=0.35$)
LME-drift	NI ($p<0.001$)	NI ($p=0.006$)	NI ($p=0.024$)
V2-perQ	NI ($p<0.001$)	NI ($p<0.001$)	NI ($p<0.001$)

The primary $\delta=0.05$ margin is permissive relative to observed RAG accuracy. At $\delta=0.03$, non-inferiority remains on the two larger benchmarks; at $\delta=0.02$, it remains on LME-drift and V2-perQ. The small LongMemEval split does not support the tighter claims.

Table 6: Held-out normalized-utility accounting. Outcome rates are rounded; paired differences use unrounded question-level scores.

Benchmark	Policy	Correct	Wrong	Abstain	U/item	Paired ΔU [95% CI]
LongMemEval s	RAG	.292	.245	.463	.670	–
	hybrid	.286	.224	.490	.686	.018 [−.015, .051]
LME-drift	RAG	.271	.259	.470	.655	–
	hybrid	.284	.218	.498	.691	.034 [.009, .059]
V2-perQ	RAG	.248	.271	.481	.641	–
	hybrid	.278	.222	.500	.686	.047 [.012, .082]

Utility is linear: the difference of mean normalized utilities equals the mean paired difference when computed from the unrounded records. Minor discrepancies obtained from the displayed rates are rounding only. For each question, all three model outcomes remain grouped before averaging.

Table 7: Held-out V2-perQ ablation contrasts ($n=225$).

Contrast	ΔAQ [95% CI]	ΔSF [95% CI]	AQ p
decay-only – RAG	.038 [.008, .068]	−.059 [−.091, −.027]	.013
gate-only – RAG	−.058 [−.089, −.027]	−.114 [−.147, −.081]	< .001
hybrid – decay-only	−.006 [−.031, .019]	−.022 [−.051, .007]	.64
hybrid – gate-only	.090 [.057, .123]	.033 [.001, .065]	< .001

Table 8: Held-out confabulation on 15 LongMemEval trap items. Each item is evaluated by three models. CF is the fraction of questions receiving a committed answer from at least one model; intervals are exact 95% Clopper–Pearson intervals.

Policy	CF	95% CI
raw context	.73	[.45, .92]
vanilla RAG	.67	[.38, .88]
hybrid	.40	[.16, .68]
decay-only	.60	[.32, .84]
gate-only	.27	[.08, .55]
full layer	.20	[.04, .48]

The denominator is 15 question-level binary outcomes per policy, derived from 45 item-model responses. A question counts as confabulated when any of its three model responses commits to an answer. This definition matches the table’s integer rates: 11/15, 10/15, 6/15, 9/15, 4/15, and 3/15.

Each sensitivity resplit reconstructs tuning and test folds at the original sizes, retunes thresholds on the tuning fold, and evaluates normalized utility on the test fold. The winner is the highest-utility policy among raw context, RAG, hybrid, full layer, Chronofy, and TEPA at $(c_{\text{wrong}}, c_{\text{abstain}})=(5, 0.1)$.

Factorial and mechanism checks

The LongMemEval factorial crosses provenance estimation (P) with lifecycle storage (L). Full-pool answer quality is 0.15 for P-/L-, 0.21 for P+/L-, 0.15 for P-/L+, and 0.16 for P+/L+. The provenance main effect is +0.034 ($p=0.013$) and the lifecycle main effect is −0.024 ($p=0.053$). The held-out provenance effect is +0.029 (CI [−0.009, 0.066], $p=0.34$); the exploratory temporal split yields +0.037 (CI [0.005, 0.069], $p=0.021$).

The synthetic generator defines truth through an event log. For each active entity-attribute, a value changes per session

with probability $p_{\text{drift}} \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$; 20% of source labels are permuted, using seed 12345. At the full-layer operating point, state resolution is 0.83 and false-premise resistance is 0.71. On 90 traps per policy, confabulation is 0.14 versus 0.74 for RAG (two-sided Fisher exact $p < 10^{-9}$). This is an operating-point benefit, not full risk-coverage dominance: at coverage 0.30, accuracy on answered items is 0.49 for full layer, 0.53 for hybrid, 0.50 for RAG, and 0.52 for raw context.

Robustness and scaling

Long context. The 47 V2-perQ histories above 150K input tokens have mean context length 178K. Hybrid answer quality is 0.23 versus RAG’s 0.18, while stale-fact rate is 0.43 versus 0.58. These subset estimates are exploratory.

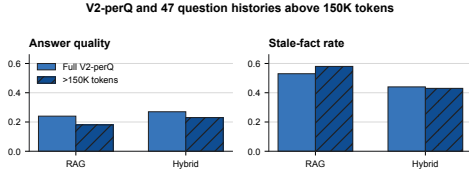


Figure 4: Full V2-perQ and exploratory > 150K-token subset results.

Stale-evidence density. Splitting V2-perQ histories at the median proportion of retrieved passages contradicted by later evidence, hybrid-minus-RAG AQ is +0.011 (CI $[-0.024, 0.046]$) in the low-density half and +0.053 (CI $[0.018, 0.088]$) in the high-density half. This exploratory interaction is consistent with a stale-evidence mechanism but does not isolate horizon.

Provenance corruption. Hybrid answer quality changes from 0.27 without corruption to 0.26 at 10% random noise, 0.23 at 20%, and 0.20 at 50%. It is 0.26 with 25% missing provenance, 0.25 with 50% missing, 0.26 under a two-session delay, and 0.24 with 15% contradictory provenance. Each condition evaluates all 597 LongMemEval item-model records over five random corruption seeds; AQ varies by at most 0.02 across seeds at each level. Corruption is applied to the belief store before answering while questions remain fixed. This supports graceful degradation under modest missingness and noise, with larger losses under severe corruption.

Drift frequency. On the synthetic sweep, full-layer state resolution changes from 0.91 at 0.1 changes/session to 0.71 at 1.0; RAG changes from 0.72 to 0.15. The absolute advantage therefore widens as drift becomes more frequent.

Efficiency. Relative to RAG, decay-only adds 11ms and 82 tokens per query, hybrid adds 23ms and 142 tokens, and the full layer adds 47ms and 218 tokens. The belief store was tested to 50K beliefs with 8ms p95 lookup latency. Raw-context replay averages 12K, 34K, and 47K input tokens across the three regimes, reaches 215K tokens at maximum, and costs 3–8 $\times$ more than RAG. No truncation or context-window failure occurred.

Citation fidelity and failure analysis

Two annotators independently reviewed paired hybrid and RAG answers for 102 V2-perQ held-out questions sampled with seed 99. Policy labels were hidden and presentation order randomized. They labeled each cited passage as direct support, partial support, or irrelevant. The 247 hybrid citations are 78%, 14%, and 8% in these categories; the 231 RAG citations are 71%, 18%, and 11%. Agreement is $\kappa=0.74$; discussion resolves disagreements.

On LongMemEval s, 597 hybrid item-model evaluations contain seven false revocations (1.2%), twelve missed updates (2.0%), eighteen excess abstentions (3.0%), and three semantic re-entry cases (0.5%). On V2-perQ, 1,353 evaluations contain nineteen (1.4%), thirty-one (2.3%), forty-two (3.1%), and eight (0.6%), respectively. A status check catches two of three LongMemEval re-entry cases.

We randomly sampled 50 LongMemEval items where hybrid committed an incorrect answer on at least one model. Mutually exclusive primary-cause labels attribute 38 (76%) to stale evidence, eight (16%) to retrieval misses, and four (8%) to reasoning with correct evidence. Thus SF captures the dominant audited failure mode on this population, but it is not exclusive to stale evidence.

Connections to recent temporal-memory methods

TEPA studies explicit stale-memory revocation [16]; STALE similarly focuses on recognizing invalid memory [14]; Chronofy introduces temporal-logical decay for time-aware retrieval [18]; MemTX studies transactional belief commit [17]; and MemOps evaluates operation-level memory behavior [15]. Memora and recent rollback work further highlight the need to recognize obsolete state and repair bad memory changes [13, 33]. Our framework places revocation in the lifecycle factor and decay in retrieval, then measures their separate effects.

Chronofy, TEPA, and STALE are implemented as paper-inspired, training-free reductions because their official systems are not compatible with our inference-only evaluation. We do not claim that our reductions establish the methods’ intrinsic limits. The Chronofy reduction has a +0.017 answer-quality point estimate over RAG on LongMemEval ($p=0.47$), close in magnitude to its reported +0.009 edge, but this serves only as a fidelity check across different evaluations.

LongMemEval-V2 reports 72.5% accuracy for its strongest active coding-agent method and 48.5% for its strongest RAG-style baseline [23]. These are not empirical competitors to V2-perQ. The official protocol tests trajectory discovery within a large haystack, whereas V2-perQ provides the question-generating trajectory and tests temporal-validity policies within it. Direct numerical comparison would be misleading. Likewise, our simplified prior-method rows test controlled mechanisms, not the complete published systems. Chronofy’s multi-stage pipeline, TEPA’s conflict graph, and STALE’s adaptive threshold may behave differently.

Reproducibility details

Models and prompts. The frozen panel contains OpenAI GPT-5.6 Terra, Anthropic Claude Sonnet 5, and Amazon Nova Premier. Calls used the Amazon Bedrock Runtime Converse API in us-east-1, July 15–18, 2026, temperature zero, and max_tokens=2048. The recorded inference profile IDs are us.openai.gpt-5.6-terra, us.anthropic.claude-sonnet-5, and us.amazon.nova-premier-v1:0. Claude thinking was explicitly disabled; the other two calls did not request a reasoning mode. The answer prompt is:

You are answering questions about a user’s history. Use ONLY the evidence passages below. If no passage supports a confident answer, respond “I cannot answer this based on available evidence.” Cite [passage_id] for each claim.

Selection and inference. Gating uses $\tau \in \{0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70\}$ and decay half-life $\lambda \in \{5, 10, 20, 50, 100\}$ sessions. The tuning fold selects the pair that maximizes normalized utility. Confidence intervals use 10,000 question-clustered BCa bootstrap resamples, keeping the model panel together. LongMemEval uses question-ID hash salt fmts04-heldout-v1; LME-drift uses a session-stratified split with seed 42; V2-perQ uses trajectory-ID hash salt fmts04-v2-v1.

RAG baseline. Amazon Titan Text Embeddings V2 (amazon.titan-embed-text-v2:0, 1024 dimensions) embeds each turn or trajectory action. Multi-paragraph turns are split at paragraph boundaries, capped at 512 tokens with 64-token overlap. A FAISS flat inner-product index retrieves $k=10$ passages by cosine similarity. Passages are appended in score order within a 4096-token evidence budget; lower-ranked overflow is dropped, and similarity ties favor recency. There is no reranker. Decay multiplies cosine similarity by the temporal factor below; gating filters ineligible lifecycle states before scoring. All other parameters are shared.

Scoring and conflicts. For belief b ,

$$\text{score}(b, q) = \text{relevance}(b, q) \text{reliability}(\text{source}(b)) \times \exp\left(-\ln 2 \frac{\text{age}(b)}{\lambda}\right).$$

Reliability multiplies the source-mention fraction by a recency weight. Structured beliefs conflict on entity-attribute key with unequal values; free-text beliefs conflict when model-judged NLI contradiction exceeds 0.7. Specifically, source reliability is the source’s fraction of all mentions, multiplied by $\exp(-\ln 2 d/\lambda)$ for d sessions since its latest mention, then normalized to $[0, 1]$ over sources in the store. The NLI judge receives two statements about the same entity and returns only a contradiction score in $[0, 1]$; it uses the same frozen model as the answering condition. ECE (0.083) and Brier score (0.21) are computed on the 199 answerable LongMemEval items,

using top-evidence reliability as predicted correctness and ten equal-width ECE bins.

Limitations

The evaluations cover factual drift in text-based memory; they do not cover visual, multimodal, or code-execution memory. V2-perQ uses oracle trajectory selection rather than the official multi-trajectory haystack and processes only text, so its results do not measure trajectory retrieval or screenshot-dependent reasoning. The LongMemEval s split has limited power for small effects. Its temporal split is exploratory. The paper-inspired reductions may differ from official implementations. Provenance estimation is lexical and slightly overconfident (ECE 0.083, Brier 0.21); learned provenance may behave differently. SF includes non-stale committed errors, although stale evidence dominates the audited sample. The $> 150\text{K}$ and stale-density analyses are exploratory, and cross-benchmark trends do not isolate horizon. We do not evaluate V2’s active coding-agent baselines. Finally, the hybrid does not dominate raw context on every benchmark: its V2-perQ AQ difference is 0.010 with CI $[-0.019, 0.039]$.